

Assignment 02
MGT 3202: Engineering Management
Summer 2025-2026

Total Marks: 15; Submission Deadline: Monday, 27th July 2026

Answer all the following questions:

Question 01: State whether the following statements are True or False. If a statement is False, write the correct statement.

- a) Effective capacity is generally lower than design capacity.
- b) A company may achieve high efficiency while operating at relatively low utilization.
- c) Capacity planning decisions usually involve long-term commitment of organizational resources.
- d) Increasing preventive maintenance can improve effective capacity without changing design capacity.
- e) If production volume is below the break-even point, the company will incur a loss.
- f) Increasing the selling price decreases the break-even quantity, assuming all other factors remain unchanged.
- g) In make-or-buy analysis, the indifference point is the production quantity at which both alternatives have the same total cost.
- h) If annual demand exceeds the indifference quantity, manufacturing in-house becomes more economical than outsourcing.
- i) Improving machine efficiency increases the effective processing capacity of each machine.
- j) Actual output can exceed effective capacity under normal operating conditions.

Question 02: Crystal Ceramics Ltd. manufactures ceramic floor tiles. Under ideal operating conditions, one production line is capable of producing 900 tiles per day. However, after considering routine maintenance, tea breaks, and product changeovers, management estimates that the line can realistically produce 780 tiles per day. Last Tuesday, the production line completed 702 tiles.

- a) Calculate the Efficiency.
- b) Calculate the Utilization.
- c) Briefly comment on the company's operating performance based on the above measures.

Question 03: Prime Packaging Ltd. plans to manufacture three biodegradable food containers next year. The forecasted annual demand and standard processing time for each product are given below.

Product	Forecasted Annual Demand (Units)	Standard Processing Time (Minutes/Unit)
A	4,000	12
B	2,800	25
C	3,500	18

Each machine operates 8 hours per day and 250 days per year. The production system is expected to operate at an average efficiency of 85%.

- a) Calculate the total annual processing time required.
- b) Determine the effective annual processing capacity of one machine.
- c) Calculate the minimum number of machines required to satisfy the forecasted demand.

Question 04: Nova Sports Ltd. plans to manufacture carbon-fiber bicycle helmets. The project requires an annual fixed operating cost of \$420,000, while the estimated variable manufacturing cost is \$35 per helmet. Management expects annual sales of 25,000 helmets and requires the project to generate an annual profit of \$280,000. Determine the minimum selling price per helmet required to achieve the target profit.

Question 05: Eco Cool Ltd. plans to introduce a new energy-efficient ceiling fan. To establish the production facility, the company expects an annual fixed operating cost of \$360,000. The estimated variable manufacturing cost is \$48 per unit. Market research suggests that approximately 18,000 units can be sold annually. Management is considering different pricing and profit targets before launching the product.

- Develop the Total Cost, Total Revenue, and Profit functions.
- If the selling price is \$84 per unit, determine the break-even quantity.
- If the company sells 18,000 units, calculate the annual profit.
- How many units must be sold to earn an annual profit of \$180,000?
- If the company expects to sell only 12,000 units but still wants to earn an annual profit of \$180,000, determine the minimum selling price per unit.

Question 06: Med-Care Instruments Ltd. currently purchases plastic syringe casings from an external supplier for \$22 per unit. The engineering department has proposed manufacturing the casings internally. If approved, the company must purchase specialized molding equipment that will result in an annual fixed operating cost of \$480,000. The estimated variable manufacturing cost will be \$13 per casing. Forecasted annual demand is 60,000 casings.

- Determine the annual indifference quantity between manufacturing in-house and outsourcing.
- Based on the forecasted annual demand, recommend whether the company should manufacture internally or continue outsourcing. Support your answer with calculations.

Question 07: Aqua Life Ltd. plans to manufacture reusable stainless-steel water bottles. Establishing the production facility requires an annual fixed operating cost of \$300,000, while the estimated variable manufacturing cost is \$20 per bottle. Management is evaluating several business scenarios before approving the investment.

- Develop the Total Cost, Total Revenue, and Profit functions.
- If the selling price is \$45 per bottle, determine the break-even quantity.
- Calculate the annual profit if the company sells 22,000 bottles.
- Determine the number of bottles that must be sold to earn an annual profit of \$200,000.
- If market research indicates annual sales of only 18,000 bottles, determine the minimum selling price per bottle required to achieve an annual profit of \$200,000.

Question 08: Aqua Flow Ltd. manufactures domestic water pumps. The assembly department currently has a design capacity of 840 pumps per day, an effective capacity of 760 pumps per day, and an actual output of 684 pumps per day. Management is considering purchasing an additional assembly line. If the investment is made, the design capacity is expected to increase to 1,100 pumps per day, the effective capacity to 980 pumps per day, and the actual daily output to 910 pumps.

- Calculate the current Efficiency and Utilization.
- Calculate the projected Efficiency and Utilization after expansion.
- Based on your analysis, discuss whether the proposed investment improves operational performance.

Question 09: Nova Textile Ltd. manufactures industrial safety uniforms. Under ideal operating conditions, the sewing department can produce 1,250 uniforms per day. After considering routine maintenance, shift changes, and quality inspections, management estimates an effective capacity of 1,080 uniforms per day. On a particular day, the department completed 972 uniforms.

- Calculate the Efficiency.
- Calculate the Utilization.
- Which performance measure is higher? Briefly explain why.

Question 10: Green Farm Equipment Ltd. plans to manufacture agricultural irrigation pumps. Each machine operates 7.5 hours per day, 300 days per year, with an expected operating efficiency of 75%.

Product	Forecasted Annual Demand (Units)	Standard Processing Time (Minutes/Unit)
Pump A	1,800	40
Pump B	2,400	25
Pump C	1,200	55
Pump D	3,000	15

- Calculate the total annual processing time required.
- Determine the effective annual capacity of one machine.
- Determine the minimum number of machines required.

Question 11: Bright Home Ltd. plans to manufacture smart LED table lamps. Annual fixed operating cost is \$520,000, variable manufacturing cost is \$32 per lamp, and the planned selling price is \$68 per lamp.

- Develop the Total Cost, Total Revenue, and Profit functions.
- Determine the annual break-even quantity.
- Calculate the annual profit if 24,000 lamps are sold.
- Determine the annual sales quantity required to earn \$260,000 profit.
- If annual sales are expected to be only 16,000 lamps, determine the minimum selling price required to achieve the same profit target.

Question 12: Smart Agro Ltd. currently purchases plastic fertilizer containers from a supplier at \$14 per container. Manufacturing the containers internally would require an annual fixed operating cost of \$650,000, while the variable manufacturing cost would be \$6 per container. Forecasted annual demand is 95,000 containers.

- Calculate the annual indifference quantity.
- Determine the total annual cost of each alternative.
- Recommend the better alternative with proper justification.

Question 13: Fresh Pack Ltd. manufactures disposable food trays. Forecasted annual demand is 480,000 trays. Each tray requires 18 seconds of machine processing time. Each machine operates 8 hours per day, 275 days per year, and the expected operating efficiency is 85%. An external supplier offers to supply the trays for \$0.75 per tray. Internal production would require an annual fixed cost of \$180,000 and a variable cost of \$0.28 per tray.

- Calculate the minimum number of machines required.
- Determine the annual indifference quantity.
- Based on the forecasted annual demand, recommend whether FreshPack Ltd. should manufacture internally or purchase from the supplier.

Question 14: Nova Medical Ltd. plans to manufacture digital thermometers. The estimated annual fixed cost is \$900,000, and the variable manufacturing cost is \$18 per thermometer.

- Develop the Total Cost, Total Revenue, and Profit functions.
- If the selling price is \$42, determine the annual break-even quantity.
- Calculate the annual profit if 60,000 thermometers are sold.
- Determine the sales quantity required to earn \$450,000 profit.
- If annual sales are expected to be only 40,000 thermometers, determine the minimum selling price required to achieve the target profit.

Question 15: Elite Furniture Ltd. is considering two production systems for manufacturing office desks. Each desk will be sold for \$72.

Alternative	Annual Fixed Cost	Variable Cost per Unit
System A	\$480,000	\$42
System B	\$720,000	\$30

- Develop the profit function for each production system.
- Determine the indifference quantity between the two alternatives.
- If expected annual demand is 28,000 desks, recommend the better production system with proper justification.

Answers to the Q. No - 01

a) True

b) True

c) True

d) True

e) True

f) True

g) True

h) True

i) True

j) ~~True~~ False $\Rightarrow$ Actual output normally can not exceed effective capacity under normal operating conditions.

Answers to the Q. No - 2

a) We know,

$$\begin{aligned}\text{Efficiency} &= \frac{\text{Actual Output}}{\text{Effective capacity}} \times 100 \\ &= \frac{702}{780} \times 100 \\ &= 90\%\end{aligned}$$

Ans.

b) We know,

$$\begin{aligned}\text{Utilization} &= \frac{\text{Actual Output}}{\text{Design Capacity}} \times 100 \\ &= \frac{702}{900} \times 100 \\ &= 78\%\end{aligned}$$

Ans.

c) The production line achieved 90% of its realistic effective capacity, ~~indicating~~ indicating strong day to day performance. However, utilization is only 78%, because design capacity is higher than effective capacity. Management could improve utilization by reducing downtime, maintenance losses, changeover time and other capacity drains.

Answer to the Question No-03

a) Here,

$$\begin{aligned}\text{Total annual processing time} &= 4000 \times 12 + 2800 \times 25 + 3500 \times 18 \\ &= 181000 \text{ minutes.}\end{aligned}$$

Ans.

b) At 85% efficiency,

$$\begin{aligned}\text{Annual effective capacity} &= 8 \times 60 \times 250 \times 0.85 \\ &= 102,000\end{aligned}$$

Ans.

c) Here,

$$\begin{aligned}\text{Machines Required} &= \frac{181,000}{102,000} \\ &= 1.77 \approx 2\end{aligned}$$

Ans.

Answer to the Q. No-04

we know,

$$\begin{aligned}P &= Q(R - V) - FC \\ \Rightarrow P + FC &= Q(R - V) \\ \Rightarrow R &= \frac{P + FC}{Q} + V \\ &= \frac{280000 + 420000}{25000} + 35 \\ &= \$63\end{aligned}$$

Ans.

Given,

$$FC = 420000$$

$$P = 280000$$

$$V = 35$$

$$Q = 25000$$

$$R = ?$$

Answers to the Q. No - 05

a) We know,

$$TC = FC + vQ$$

Now,

$$P = Q(R - v) - FC$$

$$\Rightarrow P = Q(R - 48) - 360000$$

At a selling price of \$84,

$$P = 84Q - 360000$$

Given,

$$FC = \$360,000$$

$$v = 48$$

$$b) Q_{BEP} = \frac{360000}{84 - 48} \quad [\because \text{profit} = 0]$$

$$= 10,000 \text{ units}$$

d) Given,

$$\text{profit} = \$180000$$

$$Q = ?$$

Now,

$$Q = \frac{180000 + 360000}{84 - 48}$$

$$= 15000 \text{ units.}$$

e) Given,

$$Q = 18000 \text{ units.}$$

$$P = ?$$

Now,

$$P = 18000(84 - 48) - 360000$$

$$= 648000 - 360000$$

$$= 288000$$

Ans.

e) We know,

Ans.

$$P = Q(R - v) - FC$$

$$\Rightarrow R = \frac{P + FC}{Q} + v$$

$$= \frac{180000 + 360000}{12000} + 48$$

$$= \$93 \text{ per unit.} \quad \text{Ans.}$$

Given,

$$Q = 12000$$

$$P = 180000$$

Answer to the Q. No-06

a) We know,

$$\begin{aligned}Q_0 &= \frac{FC_{\text{in-house}}}{R - v} \\&= \frac{480000}{22 - 13} \text{ units} \\&= 53,333.33 \text{ units.}\end{aligned}$$

Given,

$$FC_{\text{in-house}} = 480000$$

$$R = 22$$

$$v = 13$$

Ans.

b) Given,

$$Q = 60000$$

Now,

$$\begin{aligned}TC_{\text{outsource}} &= R \times Q \\&= 22 \times 60000 \\&= \$1320000\end{aligned}$$

Again,

$$\begin{aligned}TC_{\text{in-house}} &= FC_{\text{in-house}} + vQ \\&= 480000 + (13 \times 60000) \\&= \$1260000\end{aligned}$$

$$\begin{aligned}\text{So, annual saving} &= 1320000 - 1260000 \\&= \$60000\end{aligned}$$

So, the company should manufacture internally and internal manufacturing saves \$60,000 annually.

Answer to the Q. No-07

a) Given,

$$FC = \$300,000$$

$$v = \$20$$

$$R = \$45$$

we know,

$$\begin{aligned}P &= Q(R - v) - FC \\&= Q(45 - 20) - 300000 \\&= 25Q - 300000\end{aligned}$$

$$b) \text{ Now, } Q_{BEP} = \frac{300,000}{45 - 20}$$

$$= 12000 \text{ bottles.}$$

Ans.

$$c) Q = 22,000$$

$$P = ?$$

We know,

$$P = 22000(45 - 20) - 300,000$$

$$= 250,000$$

$$d) P = \$ 200,000$$

we know,

Ans.

$$Q = ?$$

$$P = Q(R - v) - FC$$

$$\Rightarrow Q = \frac{P + FC}{R - v}$$

$$= \frac{200,000 + 300,000}{45 - 20}$$

$$= 20,000 \text{ bottles.}$$

$$e) Q = 18,000 \text{ bottles}$$

we know,

Ans.

$$R = ?$$

$$P = Q(R - v) - FC$$

$$\Rightarrow R = \frac{P + FC}{Q} + v$$

$$= \frac{200000 + 300000}{18000} + 20$$

$$= \$47.78 \text{ per bottle.}$$

Ans.

Answer to the Question No- 8

a) we know,

Given,

$$\text{Efficiency} = \frac{\text{output}}{\text{Effective capacity}} \times 100$$

$$= 90 \%$$

$$\text{Design capacity} = 840$$

$$\text{Effective capacity} = 760$$

$$\text{Actual Output} = 684$$

$$\text{Utilization} = \frac{\text{output}}{\text{Design capacity}} \times 100$$

$$= 81.43 \%$$

b) We know,

$$\begin{aligned}\text{Efficiency} &= \frac{\text{Output}}{\text{Effective Capacity}} \times 100 \\ &= \frac{910}{980} \times 100 \\ &= 92.86\% \quad \text{Ans.}\end{aligned}$$

Given,

$$\begin{aligned}\text{Design C} &= 1100 \\ \text{Effective C} &= 980 \\ \text{Output} &= 910\end{aligned}$$

$$\begin{aligned}\text{Utilization} &= \frac{\text{Actual Output}}{\text{Design Capacity}} \times 100 \\ &= \frac{910}{1100} \times 100 \\ &= 82.73\% \quad \text{Ans.}\end{aligned}$$

c)

Measure	Current	Projected
Output	684	910
Efficiency	90%	92.86%
Utilization	81.43%	82.73%

The investment improves actual output, efficiency and utilization.
Therefore the proposed investment improves operational performance.

_____ x _____
Answer to the Q. No - 09

a) Given,

$$\begin{aligned}\text{Design capacity} &= 1250 \\ \text{Effective capacity} &= 1080 \\ \text{Actual output} &= 972\end{aligned}$$

We know,

$$\begin{aligned}\text{Efficiency} &= \frac{972}{1080} \times 100 \\ &= 90\%.\end{aligned}$$

Ans.

b) We know,

$$\text{Utilization} = \frac{972}{1250} \times 100$$
$$= 77.76\%$$

c) Efficiency is higher because actual output is compared with effective capacity, which already accounts for maintenance, shift changes and inspections. So, efficiency is higher than utilization.

————— x —————
Answer to the Q. No - 10

a) Annual processing time,

$$\text{For pump A : } 1800 \times 40 = 72000 \text{ min.}$$

$$\text{For pump B : } 2400 \times 25 = 60000 \text{ "}$$

$$\text{For pump C : } 1200 \times 55 = 66000 \text{ "}$$

$$\text{For pump D : } 3000 \times 15 = 45000 \text{ "}$$

$$\therefore \text{Total processing time} = 72000 + 60000 + 66000 + 45000$$
$$= 243000 \text{ min}$$

$$\text{b) Effective capacity at 75\% efficiency} = 7.5 \times 60 \times 300 \times 0.75$$
$$= 101250$$

Ans.

$$\text{c) Machines Required} = \frac{243000}{101250}$$
$$= 2.4 \approx 3$$

Ans.

————— x —————

Answer to the Q. No-11

Given,

$$FC = \$520000$$

$$v = \$32$$

$$R = \$68$$

a) we know,

$$TC = FC + vQ$$

$$\Rightarrow TC = 520,000 + 32Q$$

$$TR = RQ$$

$$\Rightarrow TR = 68Q$$

$$P = Q(R - v) - FC$$

$$\Rightarrow P = 36Q - 520,000$$

b) Hence,

$$Q_{BEP} = \frac{520,000}{68 - 32}$$

$$= 14444.44 \text{ lamps.}$$

Ans.

c) $Q = 24000$ lamps

$$P = ?$$

$$\text{Now, } P = 36Q - 520000$$

$$= 36 \times 24000 - 520000$$

$$= 344000.$$

Ans.

d) $P = \$260,000$

$$Q = ?$$

we know,

$$Q = \frac{P + FC}{R - v}$$

$$= \frac{260000 + 520000}{68 - 32}$$

$$= 21666.67 \text{ lamps.}$$

Ans.

e) we know,

$$R = \frac{P + FC}{Q} + v$$

$$= \frac{260000 + 520000}{16000} + 32$$

$$= \$80.75 \text{ per lamp}$$

Ans.

Answer to the Q. No - 12

Given,

$$\text{Supplier Price} = \$14$$

$$FC_{\text{in-house}} = \$650,000$$

$$v = \$6$$

$$Q = 95000$$

a) we know,

$$Q_0 = \frac{FC_{\text{in-house}}}{R - v}$$

$$= \frac{650000}{14 - 6}$$

$$= 81,250 \text{ containers.}$$

Ans.

b) we know,

$$TC_{\text{Supplier}} = RQ$$

$$= 14 \times 95000$$

$$= 1330000$$

$$TC_{\text{in-house}} = FC + vQ$$

$$= 650000 + 6 \times 95000$$

$$= 1220000$$

$$\begin{aligned} \text{c) Saving} &= 1330000 - 1220000 \\ &= 110000 \end{aligned}$$

So, Manufacture the containers internally. Demand exceeds the indifference quantity and in-house production saves \$110,000 annually.

_____ x _____

Answer to the Q. No-13

a) Here,

$$\text{Machines required} = \frac{480000 \times 18}{8 \times 60 \times 60 \times 275 \times 0.85}$$
$$= 1.283 \approx 2 \text{ machines.}$$

Ans.

b) Here,

$$Q_0 = \frac{180000}{0.75 - 0.28}$$
$$= 389978.72 \text{ units.}$$

Ans.

c) Here,

$$TC_{\text{Supplier}} = 0.75 \times 180,000$$
$$= 360000$$

$$TC_{\text{Internal}} = 180000 + 0.28 \times 180000$$
$$= 180000 + 134400$$
$$= 314400$$

$$\text{Savings} = 360,000 - 314,400$$
$$= 45600$$

So, Manufacture Internally. The demand exceeds the indttern quantity and internal production saves 45600 annually.

_____ x _____

Answer to the Question No-14

Given,

$$FC = 900000$$

$$v = 18$$

a) Now,

$$TC = 900000 + 18Q$$

$$TR = RQ$$

$$\Rightarrow TR = 42Q$$

$$\text{and, } P = Q(R - v) - FC$$

$$\Rightarrow P = 24Q - 900000$$

b) Here,

$$Q_{BEP} = \frac{900000}{42 - 18}$$

$$= 37500$$

Ans.

c) Given,

$$Q = 60,000$$

$$P = ?$$

Now,

$$P = 60000(42 - 18) - 900000$$

$$= 540000 \quad \text{Ans.}$$

$$d) P = 450,000$$

$$Q = ?$$

Now,

$$Q = \frac{450000 + 900000}{42 - 18}$$

$$= 56250 \quad \text{Ans.}$$

e) We know,

$$R = \frac{P + FC}{Q} + v$$

$$= \frac{450000 + 900000}{40000} + 18$$

$$= 51.75$$

Ans.

Given,

$$Q = 40000$$

$$R = ?$$

x

Answer to the Q. No - 15

a) For system A:

$$P_A = Q(R - V_A) - FC_A$$

$$\Rightarrow P_A = Q(72 - 42) - 480000$$

$$\Rightarrow P_A = 30Q - 480000$$

For system B:

$$P_B = Q(R - V_B) - FC_B$$

$$\Rightarrow P_B = Q(72 - 30) - 720000$$

$$\Rightarrow P_B = 42Q - 720000$$

b) Here,

$$P_A = P_B$$

$$\Rightarrow 30Q - 480000 = 42Q - 720000$$

$$\Rightarrow 12Q = 240000$$

$$\Rightarrow Q = 20000$$

So,

$$P_A = 30 \times 20000 - 480000$$

$$= 120000 \quad \text{Ans.}$$

$$P_B = 42 \times 20000 - 720000$$

$$= 120000 \quad \text{Ans.}$$

c) For system A:

$$\begin{aligned} P_A &= 30(28000) - 480000 \\ &= 360000 \end{aligned}$$

For system B:

$$\begin{aligned} P_B &= 42(28000) - 720000 \\ &= 456000 \end{aligned}$$

$$\begin{aligned} \text{profit} &= P_B - P_A \\ &= 96000 \end{aligned}$$

So, system B should be selected and generates 96000 more annual profit.